

Video Transcript

Video 10: Redundant Dictionaries in Python Solution (04:47)

So, this is the exercise on student grades where we're using redundant dictionaries. And what we mean by that is we're storing the data in several different formats. So, for example, here we've got the `studentsById`; getting the student ID will give us the student name. This one `studentsByName`, given the student name, we can get their ID. Then with `gradesByStudent`, if we have the student ID we can get a list of their grades. And in this case given the `pset_id`, we can get a list of the grades that the students got on that problem set. So, we did `getGradesOnPset` where we got the `pset_id` and this gave us a list of the students. So, student one got 125. This was on problem set three, student 2 got 115, student 3, 110.

Now, what we want is we want to add put the name of the student and the grade that they got to make this more intelligible. So, here we've got the function `gradesOnPset`, and now we've got to create this from this. Now, we know we've seen dictionary comprehensions where we take an input of a dictionary and we output a transformed dictionary. So, that's exactly what we need to do. So, let's first use this since that's where we want to start. And let's call this `students`. And now we've got this dictionary. What we want to do is to loop over that and produce another dictionary. So, we know it's a dictionary comprehension. And now, we're going to do a 'for' loop over this dictionary, so over `students`. And what we want in that is we want to get the key and we want to get the points. Now, the key is the student ID, so let's call it that. So, for `student` and `points` in `students` and we must get the items. We're good. Now, what do we want to output here? Well, we're going to output... Well, we need to transform the student ID to a name.

So, given the student ID, we can get the name by using the database and we'll get `studentsById`. So, now that's got us here, we've got this, and we just need to put in the student ID and that will return to us the name. So, we've got the `student_id`, we just got it here from `students`, and that's the name. And now, we just need to put the `:points` as the value. And we've got that as well. So, that should do it. Let me put this on the next line just so you can see. So, here we are, and we need to return that. So, that's our dictionary comprehension. Let's see if we're right. Yep, executed it and here we are. So, it's fairly straightforward if you understand what you're trying to achieve. So, given this, we could transform it into this by using the `studentsById` dictionary. You

can imagine there's lots of other transformations that we'll need, and we may do an exercise on that, but this is the solution to this present exercise. Okay. Hope you enjoyed it. Bye for now.

